

Computational Geometry (2025)

Theory Questions

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Question 1 (Analysis of Marriage before Conquest () for part (a), (***) for part (b)):**

- (a) Prove that the Marriage before Conquest algorithm runs in $O(n \log h)$ time where h is the number of points on the convex hull (hint: look at the recursion tree. Observe that each time you recurse, the number of points halves).
- (b) Let $x_1 < \dots < x_h$ be the x -coordinates of the points on the upper hull and let n_i be the number of input points $p = (x, y)$ such that $x_i \leq x < x_{i+1}$, $1 \leq i < h$. Show that the upper hull computation in the Marriage before Conquest algorithm runs in time

$$O\left(\sum_{i=1}^{h-1} n_i \log\left(\frac{n}{n_i}\right)\right).$$

Question 2 (Areas inside a polygon (*)): A lens is a geometric shape that is defined as the intersection of two disks (two circles with their interior). Let P be a polygon with n vertices and L be a lens. Show that we can compute the total area of L that lies inside P in $O(n \log n)$ time.

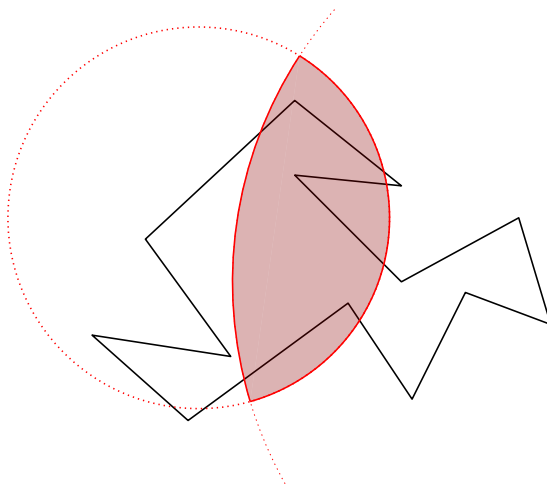


Figure 1: We want to compute the area of the part of the lens that is inside the polygon.

Question 3 (Stabbing Segments.): Let S be a set of n line segments in the plane. We say S can be *stabbed* if there exists a line ℓ that intersects all the segments in S .

- (part 1 (***)) Show that if all the line segments in S are vertical, then we can decide whether ℓ exists or not in $O(n)$ time.
- (part 2 (****)) Show that in general, where the line segments can have arbitrary orientations, in $O(n^2)$ time we can decide if such a line ℓ exists.

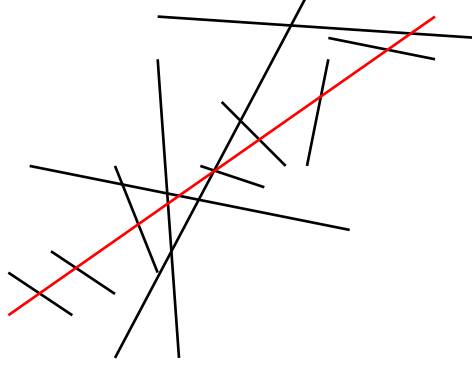


Figure 2: The red line stabs all the black line segments.

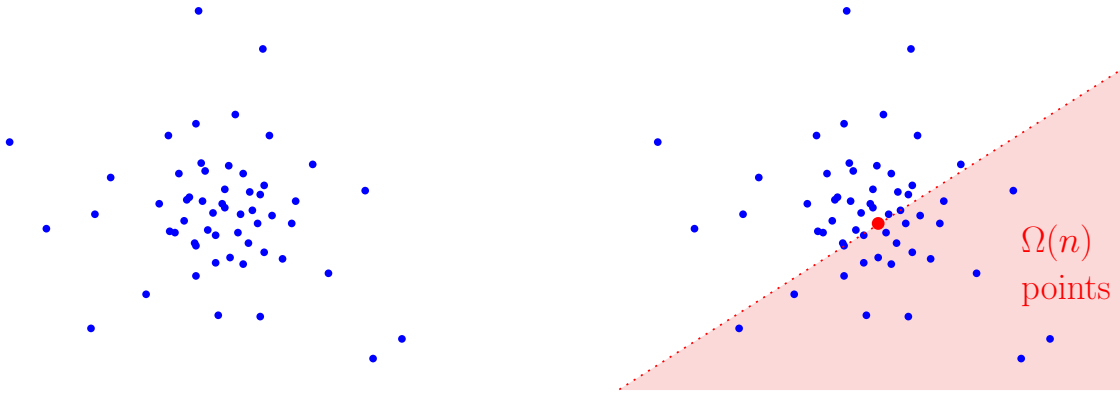


Figure 3: (left) n measurements. (right) A central point.

Question 4 (Robust Statistics: Halfspace Depth (*)**: Consider an application where we would like to estimate the location of a stationary object. The object is equipped with a sensor that constantly sends us its location, however, due to noise and other errors, each time we receive a slightly different location. We represent these locations using a point set P in the plane that includes n signals that we have received where each point represents one received signal. See Figure 3(left).

In this scenario, the points at the boundary of the point cloud (e.g., the points on the convex hull of P) represent outliers and they are way less likely to be close to the actual location of the object. Ideally, we would like to find a point that is in the “middle” of the point cloud, however, defining what is the middle of a given point set is not so straightforward.

This motivates us to define what is known as the *Tukey depth* or *Halfspace depth* of a given point q . It is defined as the minimum number of points of P contained in any halfplane that contains q . In other words, the halfspace depth of q is equal to t if and only if any line that goes through q has at least t points of P on its either side and there exists at least one line ℓ that goes through q where that are exactly t points on one side of ℓ .

Show that there always exists a point q in the plane with halfspace depth of $\Omega(n)$. For this, you can assume that there are no three points of P on the same line.

Remarks. The classical result here is that there is always a point with halfspace depth of $\frac{n}{3}$ and this can be proven using Helly's theorem. However, we have not discussed Helly's theorem in the class so for this exercise, you **cannot** use Helly's theorem.

General remarks. Expected bounds are acceptable. If you are stuck on a problem, try to review the lectures and the topics covered in the class and perhaps you suddenly remember something that helps you solve the problem.

Difficulty Notation.

- (★) This is an “easy” question, meaning, the solution requires just one idea. It does not mean that figuring out the one idea is easy but for these questions you basically need a “light bulb to turn on” in your head. These question have very short solutions; sometimes the solution can just be one line!
- (★★) This is the next stop. You need the initial idea, but even after having the itial idea, you will still have one or two more steps to go. These steps are typically much easier than getting the first initial idea but they still require some care.
- (★★★) This signifies that the problem requires a few somewhat complicated steps, or that the solution is longer than average.
- (★★★★) This is a challenging problem. It should be solvable if you spend enough time on it.
- (★★★★★) These problems are very difficult to solve. This often means that it requires you to go beyond the ideas covered in the class and be innovative.
- (★★★★★★) So you want to stay up many nights in bed, thinking about this problem?